

Sajid Ahmed

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Portfolio | Google Scholar | Dhaka, Bangladesh

Research Interests

Computer vision and multimodal foundation models; AI-enabled cyber-physical systems and sensor fusion for smart-city infrastructure; anomaly detection and forecasting on real-world streaming data.

Education

East West University — B.Sc. in Computer Science and Engineering (CSE)

2022 – 2026

CGPA: 3.70/4.00

Dhaka, Bangladesh

Thesis

IoT Water Quality Monitoring & Cybersecure Anomaly Detection — Cyber-Physical System for Smart-City Infrastructure

Python, Facebook Prophet, Isolation Forest, MQTT, ESP32

- Designed and deployed a full cyber-physical sensing system: an ESP32-based IoT node fusing pH, turbidity, TDS, and temperature streams via MQTT over TLS 1.2, with an end-to-end pipeline over 4,316 observations for forecasting (Prophet, temperature RMSE: 0.4614) and anomaly detection
- Applied Isolation Forest ($n_{\text{estimators}}=200$) for physical-world event detection (38/43 detection overlap on a real multi-sensor disturbance); end-to-end sensing-to-insight ownership directly transferable to smart-city sensor networks and digital-twin data pipelines

Publications

MSCL-DINO: A Multi-Scale Lesion-Aware DINOv2 Framework for Multi-Class Gastrointestinal Endoscopy Classification 2026

Diagnostics (MDPI) — Submitted — Sajid Ahmed*, Kabira Kalam, Nusrat Jahan Chawdhury, Samia Rahman, K. M. Safin Kamal, Ahmed Wasif Reza

- Hybrid DINOv2 Vision Transformer for GI endoscopy classification: 90.04% weighted F1 (4-class) and 95.46% weighted F1 (14-class), with a novel gradient-free explainability module

Optimized Ensemble Architecture Integrating LSTM, FFNN, and SVM for Binary Prediction Tasks 2025

IBDAP-2025 Conference, 6 pages — Sajid Ahmed, Md Al Imran*, Swabirah Iffat Binte Ameer

- Achieved 99.5% accuracy and 1.00 ROC-AUC via Optuna tuning and stratified cross-validation; stability confirmed at $p < 0.01$

Configuration Quality Failures in LLM-Generated GitHub Actions Workflows: A Comparison with Human-Validated Baseline

CCIW @ ICST — Sajid Ahmed, Samia Rahman, Tasnia Farin Ifa, Proma Chowdhury

- Empirical study of structural and semantic quality failures in LLM-generated CI/CD workflows using the BuildSonic smell taxonomy, benchmarked against expert human-written baselines

Experience

BrainiconTech — Full Stack Engineer Intern

Jun 2026

Python, Django REST Framework, React, JavaScript

Dhaka, Bangladesh

Honours and Awards

- Dean's List Scholarship (Fall 2024) — maintained 3.75+ CGPA across three consecutive semesters (35+ credits)
- Merit Scholarship (Fall 2025) — achieved 3.90+ CGPA across three consecutive semesters (35+ credits)

Skills

Computer Vision / ML / AI: DINOv2, Vision Transformers (ViT), PyTorch, Explainable AI (EigenCAM), Deepfake & Image Classification, HuggingFace Transformers, Time-Series Forecasting (Prophet), Anomaly Detection (Isolation Forest), Ensemble Methods, Feature Engineering, Hyperparameter Tuning, Model Evaluation

Embedded / Cyber-Physical Systems: ESP32, Multi-sensor Data Acquisition (pH, Turbidity, TDS, Temperature), MQTT over TLS, Signal Acquisition & Cloud Integration

Languages: Python, Java, C, HTML/CSS, PHP

Frameworks & Libraries: Scikit-learn, Pandas, NumPy, Django, React, Bootstrap, Hadoop, Spark

Databases & Tools: MySQL, SQLite, FAISS, Git, GitHub, Jira, Trello, Figma, n8n